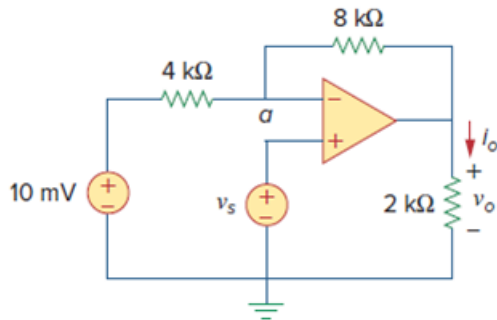


Problem

If $v_s = 8 \text{ mV}$ in the circuit of Fig. 5.41, the output voltage is:

- (a) -44 mV
- (b) -8 mV
- (c) 4 mV
- (d) 7 mV

Figure. 5.41:



Step-by-step solution

Step 1 of 1

Refer to Figure 5.41 in the textbook.

Consider that the input voltage, $v_s = 8 \text{ mV}$.

Consider that the voltages at inverting and non-inverting terminals of an ideal op amp are equal due to virtual ground.

$$\begin{aligned} a &= v_s \\ &= 8 \text{ mV} \end{aligned}$$

Apply Kirchhoff's current law at node a in the circuit.

$$\begin{aligned} \frac{a - 10 \text{ mV}}{4 \text{ k}\Omega} + \frac{a - v_o}{8 \text{ k}\Omega} &= 0 \\ \frac{8 \text{ mV} - 10 \text{ mV}}{4 \text{ k}\Omega} + \frac{8 \text{ mV} - v_o}{8 \text{ k}\Omega} &= 0 \quad (\text{since, } a = 8 \text{ mV}) \\ \frac{-v_o}{8 \text{ k}\Omega} &= \frac{2 \text{ mV}}{4 \text{ k}\Omega} - \frac{8 \text{ mV}}{8 \text{ k}\Omega} \\ \frac{-v_o}{2} &= 2 \text{ mV} - 4 \text{ mV} \\ v_o &= -2(-2 \text{ mV}) \\ &= 4 \text{ mV} \end{aligned}$$

Thus, the output voltage v_o is 4 mV .

Hence, the correct option is **(c)**.